

Tushar Chaugule

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Professional Summary

Computer Engineering undergraduate with hands-on experience building full-stack, backend, AI-enabled, real-time, and system-oriented applications. Experienced with React.js, Node.js, Express.js, FastAPI, TypeScript, Python, C++, SQL, MongoDB, REST APIs, and WebSocket, with a strong foundation in DSA, OOP, DBMS, Operating Systems, and Computer Networks.

Technical Skills

Programming: C, C++, Python, JavaScript, TypeScript, SQL

Frontend: HTML, CSS, React.js, React 19, Vite

Backend: Node.js, Express.js, FastAPI, REST APIs, WebSocket, Pydantic

Databases: MongoDB, MySQL, SQLite

AI/GenAI: NLP, RAG, TF-IDF, Multimodal AI

Systems: Embedded Systems, IoT, Three-Tier Architecture, Event-Driven Systems, System Telemetry

Tools & CS: Git, GitHub, Linux, VS Code, Postman, Pytest, Proteus, DSA, OOP, DBMS, OS, Computer Networks

Software Engineering Projects

- **KS Sentinel 2.0 — Secure Workspace OS** *React 18, Vite, Node.js, Express, MongoDB* [\[Git-Hub\]](#)
 - Built a web-based desktop workspace integrating project management, filesystem access, system telemetry, applications, virtual windows, dashboards, and notifications.
 - Implemented a three-tier architecture separating the React/Vite Web OS, Node/Express Secure Gateway, and Windows Local Agent for controlled local-machine operations.
 - Added workspace isolation, bounded filesystem roots, path validation, agent lifecycle/heartbeat, real-time system information, and automated security testing.
- **Virtual IoT Security Laboratory — End-to-End IoT Security Testbed** *Python, FastAPI, React 19, TypeScript, C, WebSocket* [\[Git-Hub\]](#)
 - Built and deployed an end-to-end software-defined IoT testbed combining virtual sensor/actuator nodes, FastAPI telemetry ingestion, real-time WebSocket streaming, and a React 19 SOC dashboard.
 - Integrated Proteus LPC2138 ARM7 simulation with C firmware and a Python UART bridge alongside virtual temperature, motion, door-lock, and HVAC devices for telemetry and command workflows.
 - Implemented authentication/RBAC gates, six deterministic detection rules, seven attack simulations, SHA-256 hash-chained auditing, quarantine controls, and forensic investigation; validated with 117/117 automated tests passing.
- **VedAI — Multimodal AI Self-Reflection Assistant** *React.js, Node.js, Express.js, Python, FastAPI* [\[Git-Hub\]](#)
 - Developed a multimodal AI application combining NLP emotion analysis, facial emotion recognition, emotion fusion, contextual guidance, and retrieval-grounded knowledge.
 - Implemented TF-IDF/cosine retrieval over 14 verified canonical verses with authentication, authorization, encrypted sensitive-data storage, layered safety detection, and automated validation.

Leadership Experience

- **Tech Team Lead** *Sep 2024 – Oct 2025*
RSCOE Mathematics Club, JSPM's RSCOE *Pune, Maharashtra*
 - Led technical execution for club activities, coordinated development tasks, and supported team members in implementing technical solutions.

Education

- **B.Tech. in Computer Engineering** *2023 – 2027*
JSPM's Rajarshi Shahu College of Engineering *Pune, Maharashtra*
 - **CGPA: 8.65/10**
 - **Minor Degree in Embedded Systems — ENTC Department:** Digital Design & Computer Organization; Microcontrollers & Laboratory; Internet of Things (IoT); Embedded Systems & Laboratory.

Certifications & Training

AWS Cloud Practitioner | **CEH Theory/Practical Training** — Completed

NPTEL Ethical Hacking — IIT Kharagpur | **NPTEL Network Security** — IIT Kharagpur | **AWS Hands-on Workshop**